LIÈGE université
Sciences Appliquées

Forecasting energy time series

ELEN0445 - Microgrids

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Overview

1. What is a forecast

2. Why forecasting? A decision-making perspective

3. Deterministic point forecasts

4. Probabilistic Forecasting

Credits

Some slides are inspired by a former version of the lecture given by Jonathan Dumas.

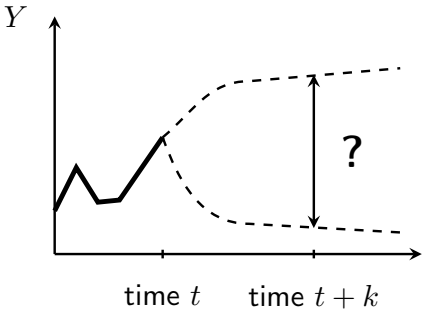
Link to the GitHub page of Jonathan

What is a forecast

Definition of the forecasting problem

We are

- at time t (e.g., at 8 am, planning the operation of the microgrid)
- interested in what will happen at time $t+k, \forall k \in \{1, 2, ..., T\}$
  - k is the **lead time**
  - T is the **forecasting horizon**
- for the random variable Y_{t+k}, (e.g. Y represents the PV production)



The most common way is to produce a **point forecast** (see next slide). It is not the only way.

Point Forecast

Definition

A single numerical value representing the "best" estimate of the future outcome based on current information $\mathcal{F}_t$.

Usually expressed as the **conditional expectation**:

$$\hat{y}_{t+k|t} = E[Y_{t+k}|\mathcal{F}_t]$$

- **Goal:** Minimize a symmetric loss function (like Mean Squared Error).
- **Output:** A single scalar.

Density Forecast

Definition

An estimate of the entire probability distribution of the random variable at time $t + k$.

Expressed as the **conditional density function**:

$$f(y_{t+k}|\mathcal{F}_t)$$

- **Goal:** Provide a complete description of uncertainty.
- **Output:** A function (e.g., a Normal distribution $\mathcal{N}(\mu, \sigma^2)$).

Quantile Forecast

Definition

The value q such that there is a probability α of the realization falling below it.

Formally defined as:

$$Q_\alpha(Y_{t+k}|\mathcal{F}_t) = \inf\{y : P(Y_{t+k} \leq y|\mathcal{F}_t) \geq \alpha\}$$

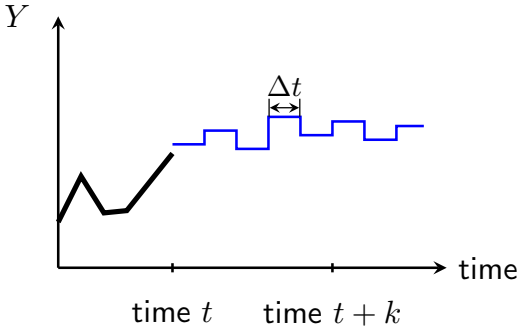
In general, we forecast several quantiles.

Key Levels:

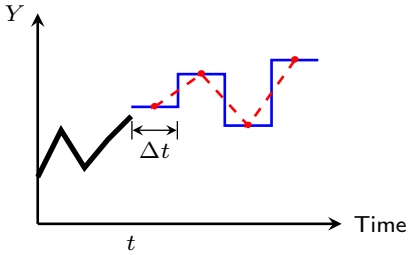
- $\alpha = 0.5$: the **median** forecast.
- $\alpha = 0.05/0.95$: the **Value at Risk** (VaR) or tail-risk boundaries.

Underlying assumption: piecewise constant forecast (Δt)

- A time step t has a duration Δt (e.g. 15 min)
- The forecasted quantity is assumed constant over this period, i.e. we are forecasting only one value (or one distribution or one set of quantiles)



Convention: Zero-Order Hold (ZOH)



- The value at the start of the interval persists until the next update.
- Mathematically:

$$Y(\tau) = \hat{y}(t),\ \forall \tau \in [t, t + \Delta t)$$

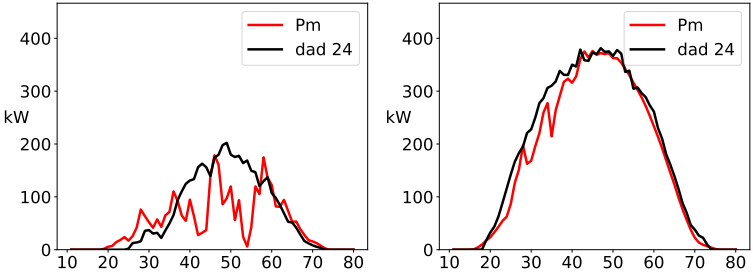
- Creates the characteristic **staircase** profile.

Example:

- if time step t corresponds to time 08:00,
- we will assume that the forecast is constant from t to $t + \Delta t$ (e.g. 8:00 to 8:15).

PV generation forecasting example I

PV point forecasts (dad 24) are computed using a neural network at noon for the next day, and the corresponding observations are in red (Pm).



PV generation forecasting example II

Inputs to the neural network: numerical weather forecasts of

- solar irradiation
- air temperature.

Reference:
J. Dumas, C. Cointe, X. Fettweis and B. Cornélusse, "Deep learning-based multi-output quantile forecasting of PV generation," 2021 IEEE Madrid PowerTech, 2021, pp. 1-6, doi: 10.1109/PowerTech46648.2021.9494976.

Why forecasting? A decision-making perspective

Operational planning: the 24h–48h horizon

- **Unit Commitment:** Deciding which local generators (e.g., Diesel GenSets, CHP) to turn ON/OFF to minimize startup costs / degradation.
- **Battery Energy Storage (BESS) Dispatch:**
  - Should we charge now from the grid (low price) or wait for the solar peak (free)?
  - Requires accurate $P_{net} = P_{load} - P_{gen}$ forecasts.
- **Demand Response (DR):** Scheduling shiftable loads (HVAC, EV charging) to align with renewable availability.

The "Cost" of Uncertainty

- Under-forecasting solar might lead to unexpected grid peak charges
- over-forecasting might lead to a depleted battery when it's needed most.

Sizing & Investment: the multi-year horizon

Long-term forecasts (10–20 years) drive the **Optimal Sizing** of assets.

- **Capital Expenditure (CAPEX):** If we over-forecast load growth, we over-invest in generation devices and batteries that remain underutilized.
- **Reliability Targets:** Using "Typical Meteorological Year" (TMY) data to ensure the microgrid can survive a 3-day "dunkelflaute" (no sun/wind).
- **Degradation Modeling:** Forecasting long-term usage patterns to predict when the BESS or fuel cell will need replacement.

Caution: when forecasting/deciding for long term, care about the **time value of money**!

Grid-connected: economic optimization

In this mode, the utility grid acts as an "infinite battery" or a safety net.

- **Objective:** Profit Maximization / Bill Reduction.
- **Forecast Impact:**
  - Errors lead to **sub-optimal trading** (e.g., buying power at peak prices).
  - Deviation from "Day-Ahead" commitments may result in **imbalance penalties** from the utility (only for large microgrids for those who have to do these commitments)
- **Safety Margin:** Low. The grid absorbs your volatility.

Key Metric

Focus on **MAPE** and **Daily Operational Cost**. A 10% error might cost $500 in lost savings, but the lights stay on.

Islanded mode: surviving

Whether in a remote area or because of a grid outage, no safety net.

- **Objective:** System stability and load shedding minimization.
- **Forecast Impact:**
  - **Under-forecasting Load:** Can lead to a sudden frequency drop and a total system blackout.
  - **Over-forecasting Solar:** May cause the battery to hit "Zero State of Charge" (0% SoC) at 2 AM.
- **Safety Margin:** High. You must use **Probabilistic Quantiles** (e.g., $q_{0.95}$) to ensure reserves are always available.

Critical Change

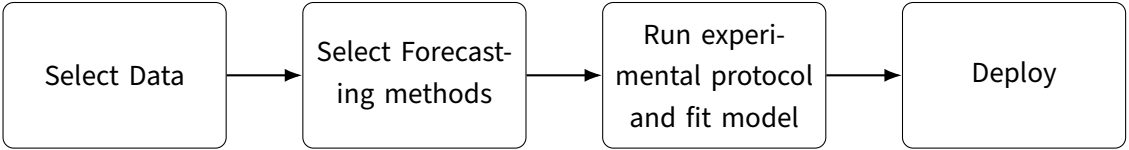
In Islanded mode, we care more about the **Lower Bound** of generation and the **Upper Bound** of load to guarantee "Survivability."

Summary: Accuracy vs. Resilience

Feature	**Grid-Connected**	**Islanded (Off-Grid)**
Primary Goal	Economic Optimization	System Reliability
Error Result	Financial Loss	Blackout / Load Shedding
Forecast Type	Point (Deterministic)	Probabilistic (Uncertainty)
Risk Appetite	Moderate	Zero to Very Low
Key Resource	Market Price Signal	Battery State of Charge

Deterministic point forecasts

Procedure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Run experi-
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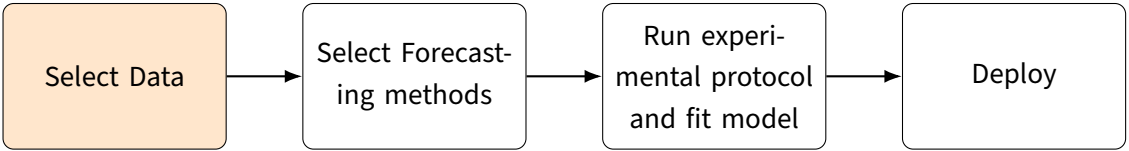
and fit model
Deploy

Procedure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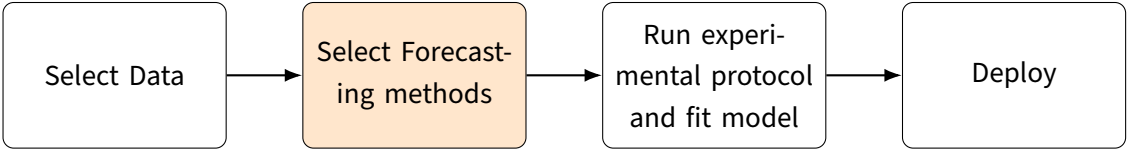
Data selection and collection

- Which data is useful for your problem?
- Which data will be avaialble when using your model? Fewer dependencies, more robustness
- Check the integrity of the data (min-max values), time-zone, averaging convention
- Fill the holes in the data and/or ensure you training procedure will not be impacted

Typical forecasting inputs

- **Weather** variables **WARNING: depends on the forecasting problem!**
  - Time series: temperature, solar irradiation, wind speed, rainfall (short term)
  - Mean/standard deviation: temperature, solar irradiation (long term)
- **Calendar** variables
  - days, hours of the days, special day (short term)
  - trend, years, months (long term)
- **Historic** values
  - at $t - 15$ min, $t - 1$ h, $t - 24$ h, $t - 7$ d, mean($t - 1$ d) (short term)
  - Mean/standard deviation: $t - 1$ week, $t - 1$ month (long term)
- **Feature engineering**
  - temperature $\times$ calendar variables
  - lagged load $\times$ temperature

Procedure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Select one or more forecasting methods

- Does a simple method yield good results?
- Do you have a lot of data?
- What is the nature of data?

Baseline model - Persistence model

- **Objective:** Establishing the "Minimum Viable Forecast."
- **Persistence Model:** The $Naive$ approach:

$$\hat{y}_{t+k} = y_t$$

Tip

If your ML model cannot beat the Persistence Model, your features are likely poorly engineered.

Machine learning models

We move from simple averages to capturing non-linear patterns:

Tree-Based **XGBoost / LightGBM**: Excellent for tabular data and handling missing weather values.

Deep Learning **LSTMs**: Captures the "memory" of power ramps in wind and solar.

Multi-step Forecasting Strategies I

To predict a horizon T, we must choose how to handle the lead time k.

1. Recursive (Iterated)

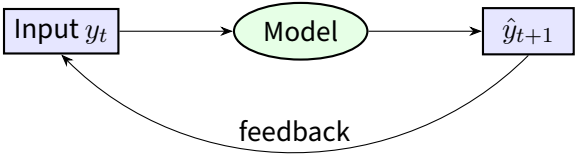
- A single model predicts $t+1$.
- The prediction $\hat{y}_{t+1}$ is fed back as an input for $t+2$.
- Risk: Error accumulation over long horizons.

2. Direct (Multi-output)

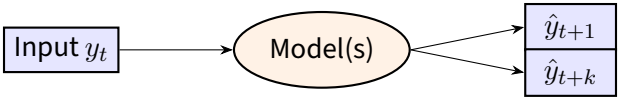
- T different models (or one multi-output NN) predict each step independently.
- No error propagation.
- Risk: Ignores temporal dependencies between outputs.

Multi-step Forecasting Strategies II

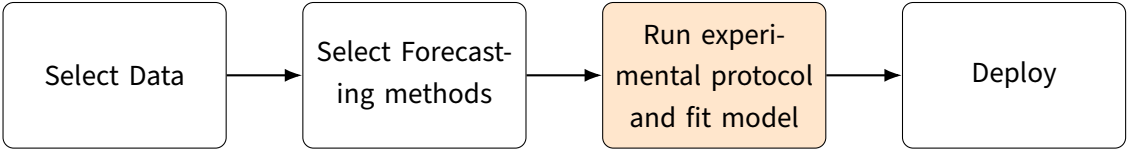
Recursive



Direct



Procedure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Define an experimental protocol to train and test your Models and **Avoid Data Leakage**

Time order is important! Shuffling your data for Cross-Validation (CV) is a critical error.

- **Data Leakage:** If the model sees tomorrow's weather to predict today's load, its accuracy is an illusion.
- **Forward Chaining CV:**
  - Fold 1: Train [Jan], Validate [Feb]
  - Fold 2: Train [Jan-Feb], Validate [Mar]
  - Fold 3: Train [Jan-Mar], Validate [Apr]

The Test Set

Always keep the final 10–15% of your most recent data (e.g., the last 3 months) completely untouched until the very end.

Avoiding Overfitting in Energy Models

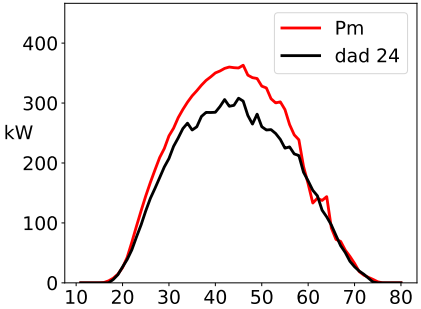
How to ensure the model generalizes to new weather patterns:

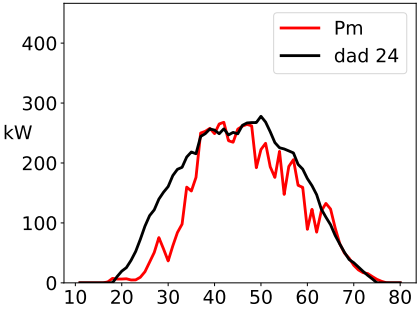
1. **Feature Selection:** Don’t throw 100 weather variables at the model. Use *Correlation Heatmaps* to pick the most impactful ones.
2. **Early Stopping:** Stop training when the **Validation Loss** starts to rise, even if Training Loss is still falling.
3. **Hyperparameter Tuning:** Use a "Grid Search" or "Bayesian Optimization" specifically within the Time-Series Split framework (e.g. **Optuna**)

Result: A model that learns the *physics* of the microgrid, not the *noise* of the sensors.

Numerical weather forecast errors

Weather forecasts errors most often drive the forecasting errors. Typical errors are amplitude errors (left below) and phase errors (right below).

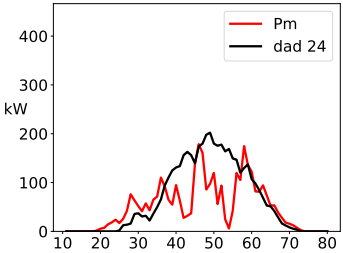




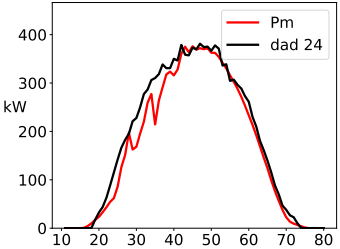
Visual inspection

Visual inspection offers significant insight into forecast trends. While it remains a strictly a qualitative assessment, it is key in the development process.

What do you think of these two? Are they good or bad?



Issued on April 3, 2020 at 12:00 for April 4, 2020.



Issued on May 4, 2020 at 12:00 for May 5, 2020.

Qualitative analysis ought to be complemented by quantitative analysis.

Let y be a realization and $\hat{y}$ a forecast.

The **forecast error** is defined by

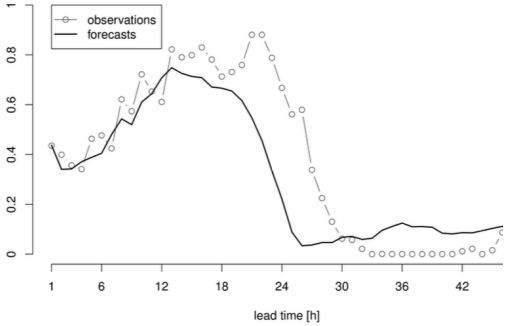
$$\varepsilon_{t+k|t} = y_{t+k|t} - \hat{y}_{t+k|t}$$

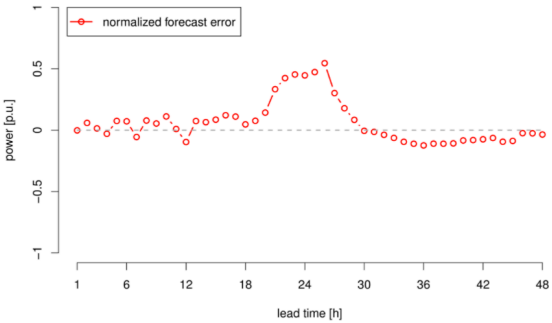
It can be **normalized**

$$\bar{\varepsilon}_{t+k|t} = \frac{y_{t+k|t} - \hat{y}_{t+k|t}}{P_n}$$

with P_n the nominal capacity (assuming a plant can generate a power between 0 and P_n).

Illustration of forecast error: a wind farm





Common error metrics or scores

Scores are used to summarize aspects of forecast accuracy.
The most common scores include, **as a function of the lead time** k:

- The bias

$$\mathbf{bias}(k) = \frac{1}{T}\sum_{t=1}^{T} \varepsilon_{t+k|t}$$

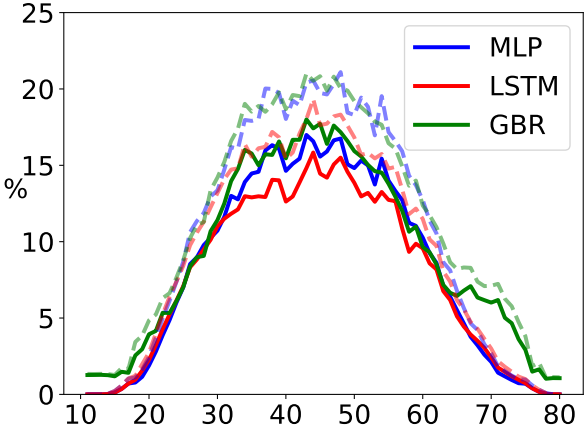
- The Mean Absolute Error (MAE) or NMAE, for the normalized version:

$$\mathbf{MAE}(k) = \frac{1}{T}\sum_{t=1}^{T} \left|\varepsilon_{t+k|t}\right|$$

- The Root Mean Squared Error (RMSE) or NRMSE, for the normalized version:

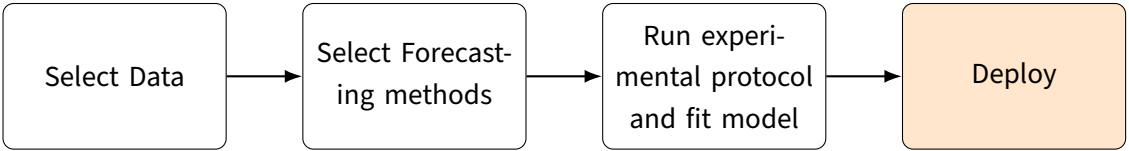
$$\mathbf{RMSE}(k) = \sqrt{\frac{1}{T}\sum_{t=1}^{T} \varepsilon_{t+k|t}^2}$$

Comparison of three models



NMAE (plain) and NRMSE (dashed) for three forecasting models.

Procedure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Deploy your model

- Write the data harvesting scripts
- Store data in a structured manner
- Use visualization tools (e.g. Grafana)
- Store your forecasts in a structured manner (do not overwrite them)
- Monitor error metrics

Probabilistic Forecasting

Handling Uncertainty

A single number is dangerous. We need **Quantile Regression**. Instead of predicting the mean, we predict the τ-th quantile:

- $q_{0.10}$: 10% chance the actual value is below this (Lower bound).
- $q_{0.50}$: The Median.
- $q_{0.90}$: 90% chance the actual value is below this (Upper bound).

Techniques: Pinball Loss optimization, Gaussian Processes, or Quantile Regression Forests.

Evaluating Uncertainty: Calibration vs. Sharpness

Calibration (Reliability)

Do your 90% intervals actually contain 90% of the data?

Sharpness

Is the interval narrow enough to be useful for decision-making?

The Scoring Rule: *Continuous Ranked Probability Score (CRPS)*

$$CRPS(F, y) = \int_{-\infty}^{\infty} (F(x) - \mathbb{1}\{x \geq y\})^2 dx$$